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EXPERIMENT 7

Experiment 7:. Random Sampling and Sampling Distribution

AIM:

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

STEPS:

1. Generate a Population:
○ Create a population of data with a specified distribution (e.g., normal distribution).
2. Random Sampling:
○ Perform random sampling from the population to create multiple samples of different sizes.
○ Compute sample statistics (mean, standard deviation, etc.) for each sample.
3. Sampling Distribution:
○ Plot histograms or density plots of sample statistics (e.g., sample means).
○ Compare the sampling distribution of the sample statistic (mean) with the population distribution.
4. Central Limit Theorem (Optional):
○ Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

CODE:

```
import numpy as np

import matplotlib.pyplot as plt

# Step 1: Generate a population (e.g., normal distribution)

population_mean = 70

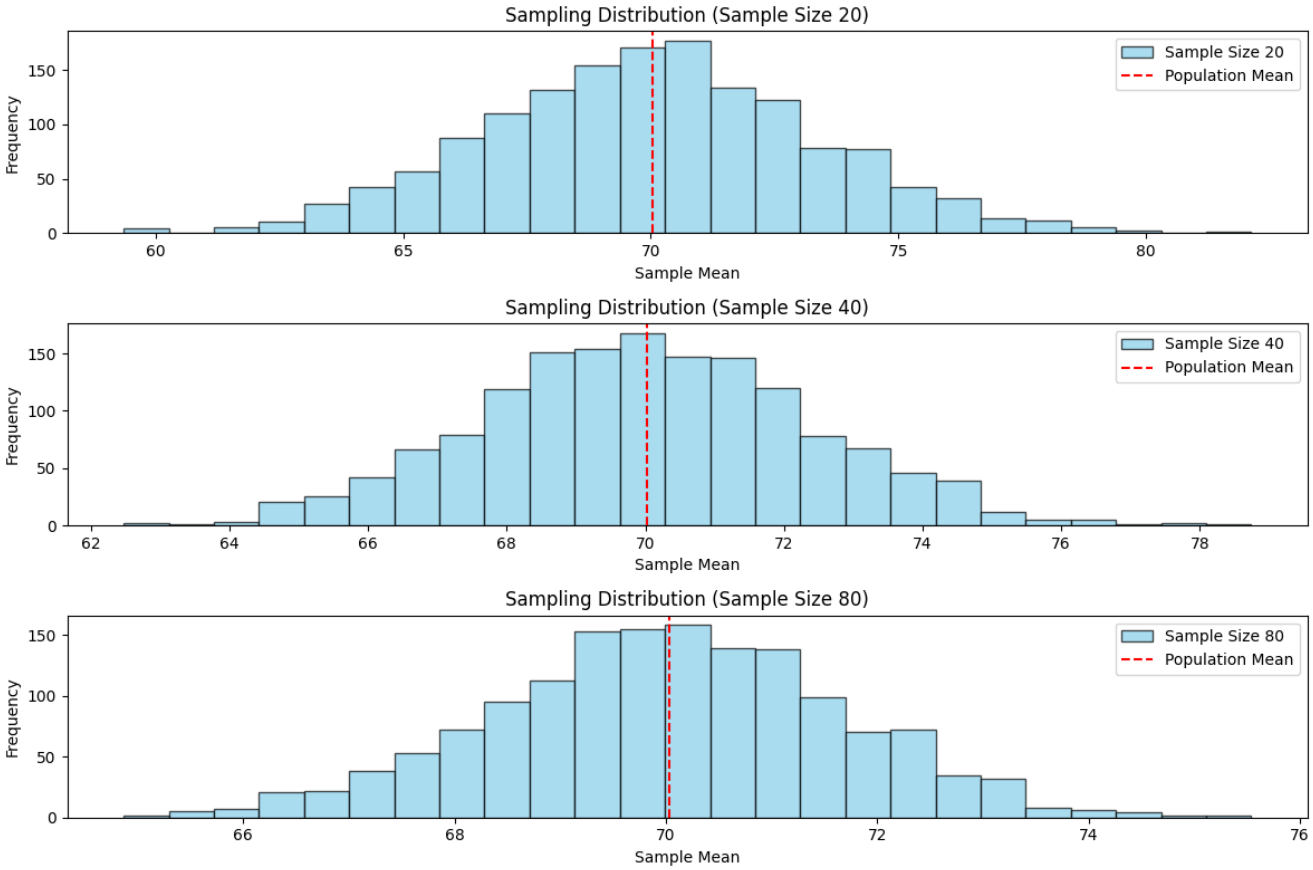
population_std = 15

population_size = 300000
```

```
population = np.random.normal(
    population_mean,
    population_std,
    population_size
)
# Step 2: Random sampling
sample_sizes = [20, 40, 80]  # different sample sizes
num_samples = 1500           # number of samples for each sample size
sample_means = {}
for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(
            population,
            size=size,
            replace=False
        )
        sample_means[size].append(np.mean(sample))
# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))
for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i + 1)
    plt.hist(
```

```
        sample_means[size],
        bins=25,
        alpha=0.7,
        color='skyblue',
        edgecolor='black',
        label=f'Sample Size {size}'
    )
    plt.axvline(
        np.mean(population),
        color='red',
        linestyle='dashed',
        linewidth=1.5,
        label='Population Mean'
    )
    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()
plt.tight_layout()
plt.show()
```

OUTPUT:



RESULT:

The distribution was generated by randomly selecting a sample and calculating the sample mean. As the sample size increased, the sample means become more concentrated around the population mean of 70.